MA388 Sabermetrics: Lesson 27

Launch Angles and Exit Velocity

```r
library(tidyverse)
library(knitr)
library(baseballr)
```

Lesson Objectives

1. Create a scatter plot showing the relationship between exit velocity, launch angle, and batting average.
2. Compare 2015 and 2017 Statcast data to investigate changes in home run rates.
3. Use wOBA as an alternative outcome measure on launch-angle/exit-velocity plots.
4. Construct histograms of launch angle and exit velocity for home runs in different years.

After seeing a rise in home runs starting in 2015, Baseball Commissioner Robert D. Manfred, Jr. assembled a commission to analyze the situation. A press release outlining the commission's findings and MLB's actions can be found here.

Let's take a look at 2017 Statcast data.

```r
# # Statcast season window for 2015 (regular + early postseason window as example)
# start_date <- as.Date("2015-04-02")
# end_date   <- as.Date("2015-11-01")
#
# # Break the range into chunks (API is happier with day- or week-sized chunks)
# date_chunks <- seq(start_date, end_date, by = "1 week")
# date_ends   <- pmin(date_chunks + 6, end_date)
#
# # Pull BIP for each chunk
# statcast_2015 <- map2_dfr(date_chunks, date_ends, ~{
#   message("Fetching: ", .x, " to ", .y)
```

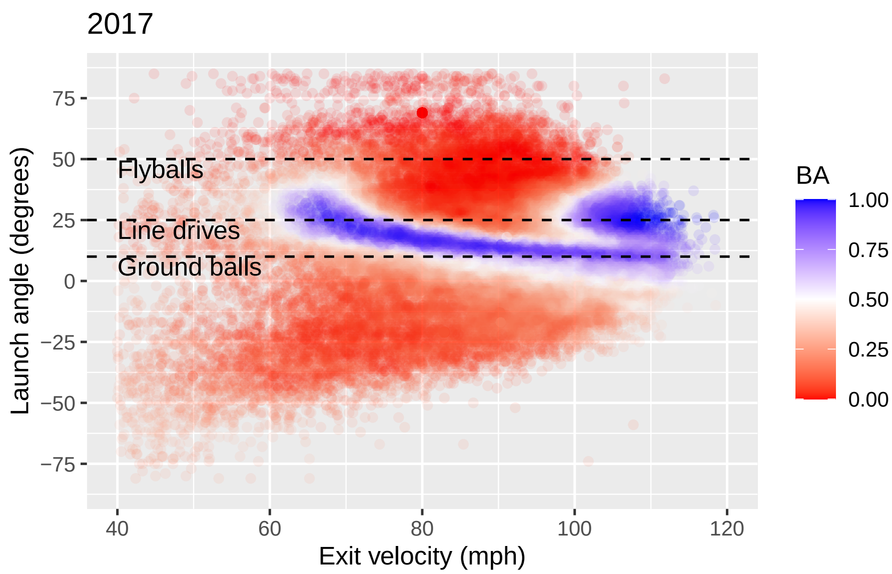
```
#   Sys.sleep(0.75)  # be polite to the endpoint
#   statcast_search(start_date = .x, end_date = .y)
# })
#
# write_csv(statcast_2015,'statcast_2015.csv')
#
# bip_2015 <-
#   statcast_2015 |>
#   filter(type=='X')
#
# write_csv(bip_2015,'statcast_2015_bip.csv')
```

Filter the data for balls in play (already did; originally 653k rows…).

```
statcast_2017_bip <- read_csv("statcast_2017_bip.csv")

statcast_bip <- statcast_2017_bip |>
  # filter(type == "X") |>
  sample_n(40000) |>
  mutate(p_throws = ifelse(p_throws == "L","P-L","P-R"),
         stand = ifelse(stand == "L","B-L","B-R"))
```

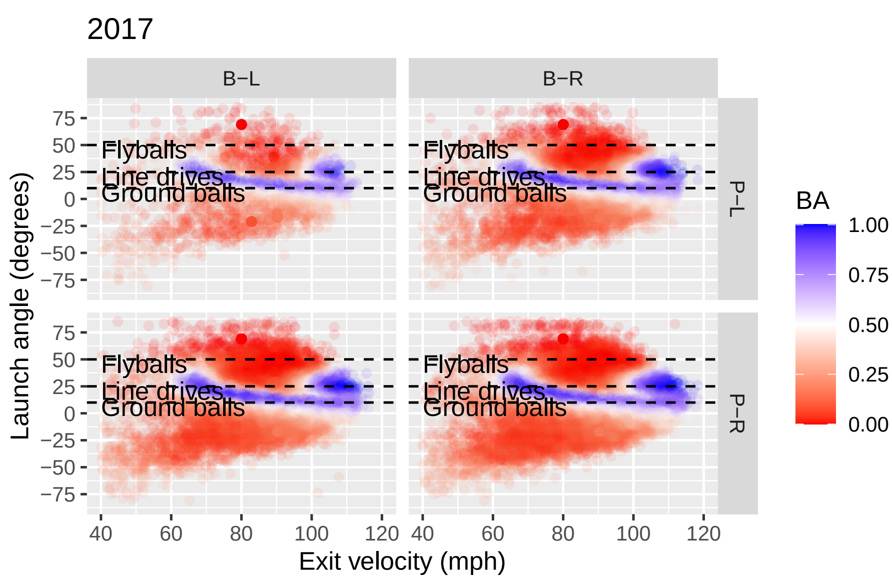
```
guidelines <- tibble(
  launch_angle = c(10, 25, 50),
  launch_speed = 40,
  label = c("Ground balls", "Line drives", "Flyballs")
)

ev_plot = statcast_bip |>
  ggplot(aes(x = launch_speed, y = launch_angle,
             color = estimated_ba_using_speedangle)) +
  geom_point(alpha = 0.1) +
  geom_hline(data = guidelines, aes(yintercept = launch_angle),
             color = "black", linetype = 2) +
  geom_text(data = guidelines,
            aes(label = label, y = launch_angle - 4),
            color = "black", hjust = "left") +
  scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=.5) +
  scale_x_continuous("Exit velocity (mph)",
                     limits = c(40, 120)) +
  scale_y_continuous("Launch angle (degrees)",
                     breaks = seq(-75, 75, 25),
                     limits = c(-85, 85)) +
  ggtitle("2017")
```

```r
ev_plot
```



Let's break this down by pitcher and batter handedness.

```r
ev_plot +
  facet_grid(p_throws ~ stand)
```

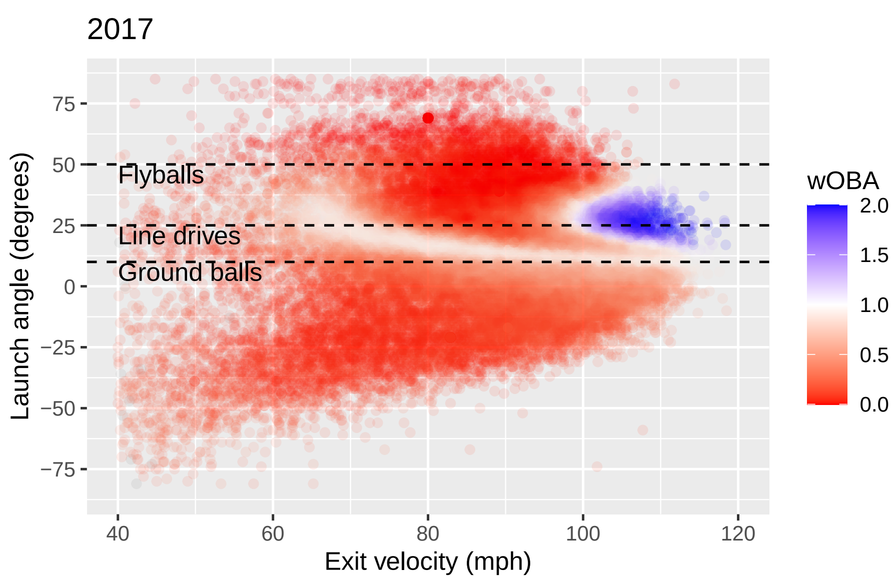


We can also look at velocity and launch angle in terms of wOBA.

From MLB.com:

> Definition: wOBA is a version of on-base percentage that accounts for how a player reached base – instead of simply considering whether a player reached base. The value for each method of reaching base is determined by how much that event is worth in relation to projected runs scored (example: a double is worth more than a single). For instance: In 2014, a home run was worth 2.101 times on base, while a walk was worth 0.69 times on base. So a player who went 1-for-4 with a home run and a walk would have a wOBA of .558 – (2.101 + 0.69 / 5 PAs).
>
> The formula: Where "factor" indicates the adjusted run expectancy of a batting event in the context of the season as a whole: (unintentional BB factor x unintentional BB + HBP factor x HBP + 1B factor x 1B + 2B factor x 2B + 3B factor x 3B + HR factor x HR)/(AB + unintentional BB + SF + HBP).

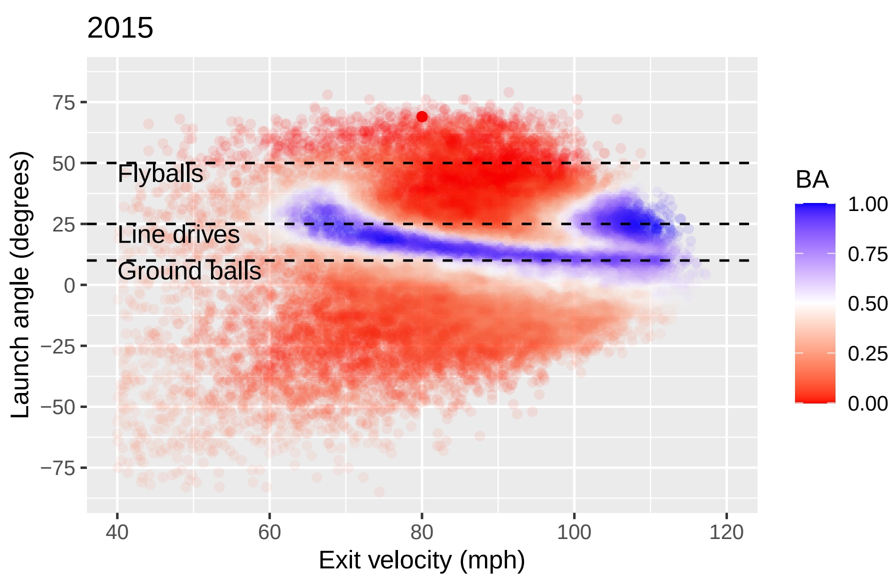
```
ev_plot +
  aes(color = estimated_woba_using_speedangle) +
    scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=1) +
  guides(color = guide_colorbar(title = "wOBA"))
```



Now let's look at the 2015 Data.

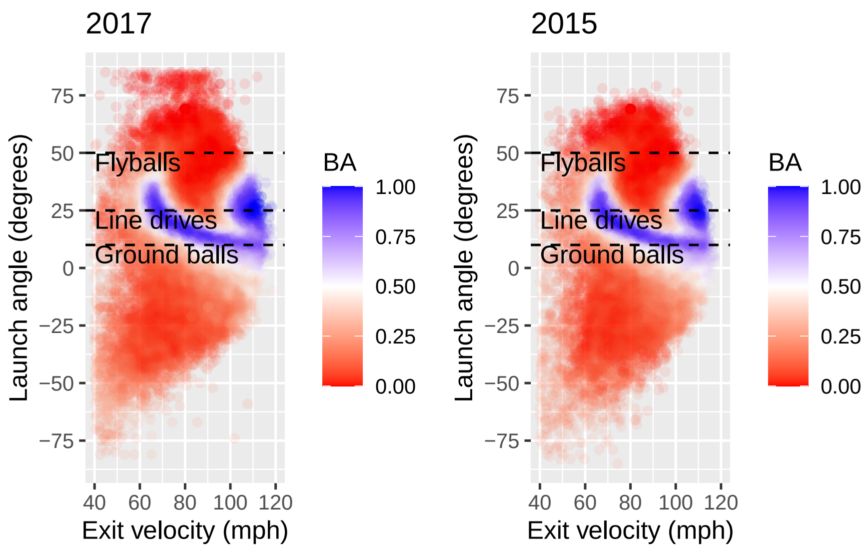
```r
statcast2015_bip <- read_csv("statcast_2015_bip.csv") |>
  sample_n(40000)
```

```r
ev_plot_2015 = statcast2015_bip |>
  ggplot(aes(x = launch_speed, y = launch_angle,
             color = estimated_ba_using_speedangle)) +
  geom_point(alpha = 0.1) +
  geom_hline(data = guidelines, aes(yintercept = launch_angle),
             color = "black", linetype = 2) +
  geom_text(data = guidelines,
            aes(label = label, y = launch_angle - 4),
            color = "black", hjust = "left") +
  scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=.5) +
  scale_x_continuous("Exit velocity (mph)",
                     limits = c(40, 120)) +
  scale_y_continuous("Launch angle (degrees)",
                     breaks = seq(-75, 75, 25),
                     limits = c(-85, 85)) +
  ggtitle("2015")

ev_plot_2015
```



Lets look at these plots together.

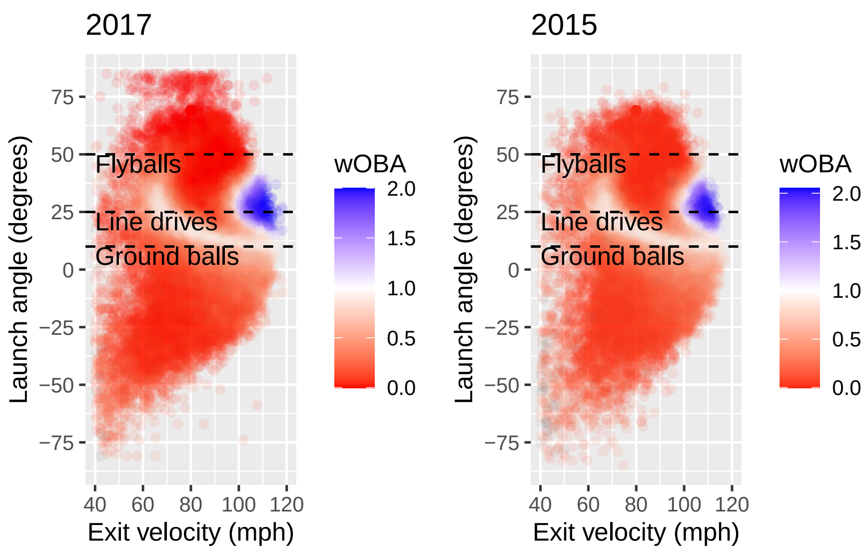
```r
library(patchwork)
a = ev_plot
b = ev_plot_2015
a+b
```



How do these two plots compare?

Let's make the same side-by-side plot for wOBA:

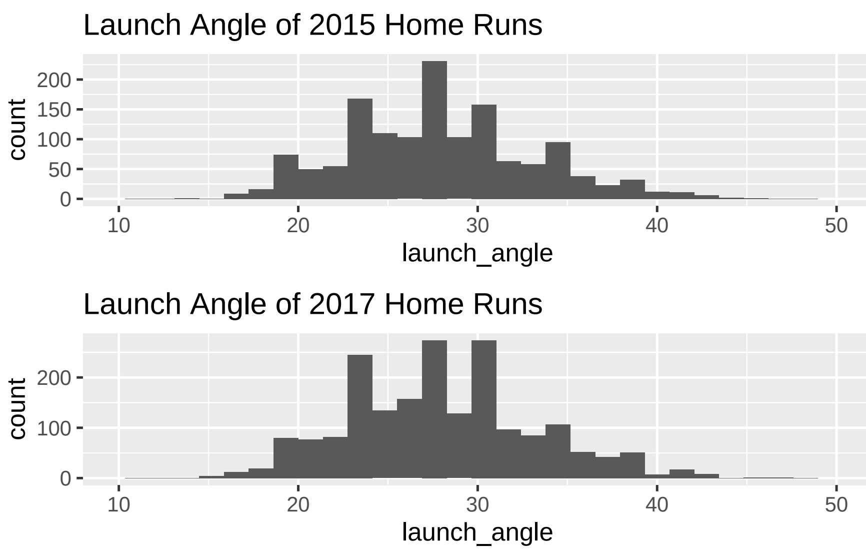
```r
c = ev_plot +
  aes(color = estimated_woba_using_speedangle) +
    scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=1) +
  guides(color = guide_colorbar(title = "wOBA"))

d = ev_plot_2015 +
  aes(color = estimated_woba_using_speedangle) +
    scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=1) +
  guides(color = guide_colorbar(title = "wOBA"))

c+d
```



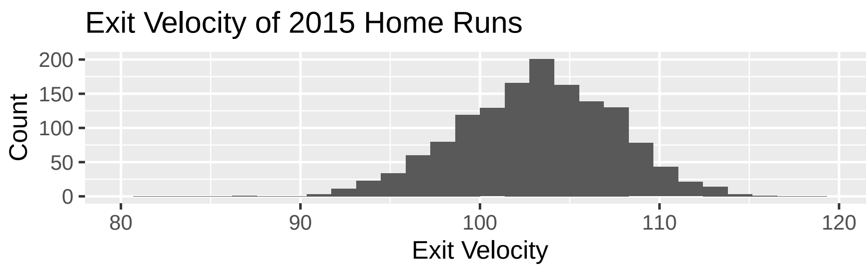
Now let's look at the launch angles and exit velocities for home runs. Construct histograms of these metrics for both 2015 and 2017.

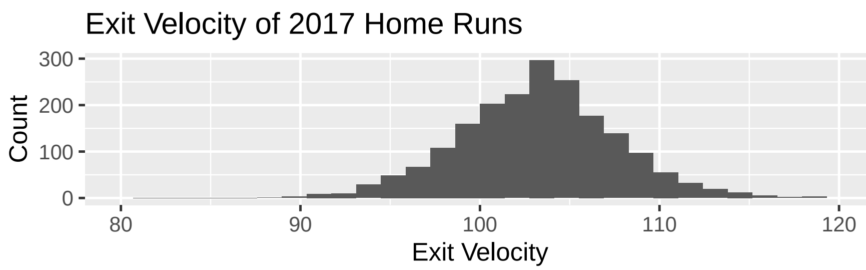
```
hr2017 <- statcast_bip |>
  filter(events == "home_run")
hr2015 <- statcast2015_bip |>
  filter(events == "home_run")
```

```
la_2015 <- hr2015 |>
  ggplot(aes(x = launch_angle)) +
  geom_histogram() +
  labs(title = "Launch Angle of 2015 Home Runs") +
  scale_x_continuous(limits = c(10,50))

la_2017 <- hr2017 |>
  ggplot(aes(x = launch_angle)) +
  geom_histogram() +
  labs(title = "Launch Angle of 2017 Home Runs") +
  scale_x_continuous(limits = c(10,50))

la_2015/la_2017
```




```
ev_2015 <- hr2015 |>
  ggplot(aes(x = launch_speed)) +
  geom_histogram() +
  labs(title = "Exit Velocity of 2015 Home Runs",
       y = "Count",
       x = "Exit Velocity") +
  xlim(80,120)

ev_2017 <- hr2017 |>
  ggplot(aes(x = launch_speed)) +
  geom_histogram() +
  labs(title = "Exit Velocity of 2017 Home Runs",
       y = "Count",
       x = "Exit Velocity") +
  xlim(80,120)

ev_2015/ev_2017
```


Exit Velocity of 2015 Home Runs
Count
200
150
100
50
0
80
90
100
110
120
Exit Velocity


Exit Velocity of 2017 Home Runs
Count
300
200
100
0
80
90
100
110
120
Exit Velocity